

Franyer López

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PROFESSIONAL SUMMARY

Reliability Engineer in the oil & gas sector (Campo Rubiales, Ecopetrol) with a background in Systems Engineering and a Specialization in Artificial Intelligence. I combine real field experience — fugitive emissions inspections, regulatory compliance — with software development: data pipelines, interactive dashboards, and AI-driven automation applied to my own field operation. Focused on building functional, well-documented, results-oriented software.

WORK EXPERIENCE

Reliability Engineer | Sep. 2022 – Present

Mecánicos Asociados (Ecopetrol contractor) · Campo Rubiales, CPF-1 and CPF-2

- I run fugitive emissions detection inspections with an OGI thermographic camera (FLIR) across 200+ clusters and the CPF-1 and CPF-2 areas, producing daily environmental compliance reports.
- I identify and track gas leaks, directly contributing to reducing emissions from the facilities.
- I designed and built an interactive dashboard for tracking per-cluster OGI inspections, digitizing a process that previously relied on manual field logging.
- I built an emissions reporting system with 3D visualization and KPIs, integrated with Ecopetrol's official GOP-F-006 format, improving field data traceability.

Freelance Developer | Jan. 2021 – Jan. 2022

JD Sistemas y Seguridad · Colombia

- Updated and maintained the organization's websites on an ongoing basis.
- Coded web sections and pages with HTML, CSS, and JavaScript, ensuring responsive design.
- Fixed structural and styling issues on existing websites, collaborating with the technical team.

FEATURED PROJECTS

Crude Oil Pipeline Supervisor · 2026 · FastAPI · PostgreSQL · PWA · Capacitor (Android) · JWT · GPS

- Full-stack system for crude oil pipeline supervision: offline PWA with GPS route logging, native Android app, backend with multi-user JWT auth (supervisor/inspector roles), and GPX export.

CPF Emissions Report · 2026 · HTML · Three.js · JavaScript · Ecopetrol GOP-F-006

- Interactive dashboard for fugitive emissions (OGI) inspections: 3D tank simulator, coverage KPIs, and severity semaphore, aligned with Ecopetrol's official GHG reporting format.

Document Automation Pipeline · 2025–2026 · Python · FastAPI · Claude Vision (Anthropic) · SQLite · Docker · Render

- Autonomous data-extraction pipeline for invoices and receipts, with a validation module combining business rules and model confidence to decide auto-approval or review, with full audit trail and rollback.

TaskFlow · 2024–2025 · React · FastAPI · PostgreSQL · JWT · PWA · CI/CD

- Full-stack Kanban project management tool with JWT authentication, installable Android PWA, deployed on Vercel, Render, and Neon with automated tests via GitHub Actions.

EDUCATION

Systems Engineering (B.Eng.)

Universidad Nacional Abierta y a Distancia (UNAD) — Graduated, 2022

Specialization in Artificial Intelligence

Corporación Universitaria Minuto de Dios (Uniminuto) — 2026

TECHNICAL SKILLS

Languages: Python (intermediate), JavaScript, HTML, CSS, SQL

Frameworks & libraries: FastAPI, Flask, React, pandas

AI & data: Claude API (Anthropic), Gemini API (Google), RAG (Retrieval-Augmented Generation), embeddings, data analysis

DevOps & tools: Git, GitHub Actions (CI/CD), Docker, Linux, PostgreSQL, SQLite, deployment on Vercel/Render/Neon

Field technical domain: OGI thermographic camera (FLIR), fugitive emissions inspection, environmental compliance (GHG)

Other: Process automation, networking & IT security, responsive web design

LANGUAGES

Spanish: Native | English: Intermediate